

Supplemental Material for “Exactly Degenerate Quantum Chaos: Non-Abelian Quantum Geometry as a Probe”

Thomas J. Wang^{1,*} and OkongOyangO²

¹*Tsinghua University, Beijing 100084, China*

²*The Pennsylvania State University,
University Park, Pennsylvania 16802, USA*

(Dated: August 17, 2026)

I. GAUGE-COVARIANT RESPONSE AND STATISTICAL ESTIMATORS

This section records the projector identities and statistical contractions used in the main text. They are stated independently of any particular parent Hamiltonian so that model-dependent exactness and gap checks can be audited separately from the response calculation.

A. Differentiation of an isolated spectral projector

Assume that $H(\lambda)$ is C^1 in a neighborhood of the base point and has a rank- D eigenprojector P separated from $Q = 1 - P$ by $\Delta > 0$. Differentiating $P^2 = P$ gives

$$P(\partial_a P)P = 0, \quad Q(\partial_a P)Q = 0,$$

so $\partial_a P$ is off diagonal in the P/Q decomposition. Differentiating $HP = E_0 P$, multiplying by Q on the left and P on the right, and using $QHP = 0$ yields

$$Q(H - E_0)Q Q(\partial_a P)P = -Q(\partial_a H)P.$$

The inverse on $Q\mathcal{H}$ gives

$$X_a = Q(\partial_a P)P = -Q[Q(H - E_0)Q]^{-1}Q(\partial_a H)P. \quad (1)$$

Hermiticity supplies the other block, $P(\partial_a P)Q = X_a^\dagger$, hence

$$\partial_a P = X_a + X_a^\dagger.$$

The norm estimate $\|X_a\| \leq \|Q(\partial_a H)P\|/\Delta$ is useful as a conditioning check, not as a prediction for the distribution of response matrix elements.

The same identities follow from the Riesz formula

$$P = -\frac{1}{2\pi i} \oint_{\Gamma} (H - z)^{-1} dz,$$

where Γ encloses only E_0 . This representation makes clear that the construction is independent of a basis chosen inside the degenerate fiber. It also states the failure mode precisely: if no contour separates the target cluster from the remaining spectrum, a fixed-rank adiabatic projector is no longer defined by this prescription.

* WangTheoPhys@outlook.com

B. Frame transformations and closed response words

In a local frame Ψ , the rectangular response is $x_a = Q(\partial_a P)\Psi$. Under $\Psi \mapsto \Psi U$,

$$x_a \mapsto x_a U, \quad x_a^\dagger x_b \mapsto U^\dagger x_a^\dagger x_b U.$$

Consequently, a trace word is independent of the frame whenever its fiber indices form a closed cycle. Examples include

$$\text{Tr}(x_a^\dagger x_b), \quad \text{Tr}(x_a^\dagger x_b x_c^\dagger x_d), \quad \text{Tr}(F_{ab} F_{cd}).$$

An individual entry of x_a is covariant and can be used inside an estimator whose contractions transform consistently. It is not, on its own, a gauge-invariant measurement. An individual entry of the connection is even less suitable because of the inhomogeneous derivative term in its gauge transformation.

For ensembles at a common base projector, one frame is fixed before sampling the tangent panel; a subsequent frame change is applied to every panel. For independent Hamiltonians, there is generally no canonical identification of their protected fibers. Such ensembles are compared through invariant scalar statistics unless a parallel-transport or polar-overlap identification is specified before the data are opened.

C. Complete complex Wick tensor

Let $(X_a)_{\alpha i}$ denote the response from fiber index i to complement index α , after centering and the registered channel whitening. Define the full entry covariance and pseudocovariance by

$$C_{ab;\alpha i, \beta j} = \mathbb{E}[(X_a)_{\alpha i} (X_b)_{\beta j}^*], \quad \Pi_{ab;\alpha i, \beta j} = \mathbb{E}[(X_a)_{\alpha i} (X_b)_{\beta j}].$$

No tensor-product decomposition of these arrays is assumed. The physical four-point tensor in these indices is

$$T_{abcd} = \frac{1}{D} \sum_{\alpha, \beta, i, j} \mathbb{E}[(X_a)_{\alpha i}^* (X_b)_{\alpha j} (X_c)_{\beta j}^* (X_d)_{\beta i}]. \quad (2)$$

Applying the three Wick pairings to Eq. (2) gives

$$\begin{aligned}
W_{abcd} = \frac{1}{D} \sum_{\alpha, \beta, i, j} & \left[C_{ba; \alpha j, \alpha i} C_{dc; \beta i, \beta j} \right. \\
& + \Pi_{ac; \alpha i, \beta j}^* \Pi_{bd; \alpha j, \beta i} \\
& \left. + C_{da; \beta i, \alpha i} C_{bc; \alpha j, \beta j} \right]. \tag{3}
\end{aligned}$$

The first and third lines pair one conjugated entry with one unconjugated entry. The middle line pairs the two conjugated entries with each other and the two unconjugated entries with each other; it is exactly the contribution controlled by the pseudocovariance. Equation (3) is therefore valid for a general zero-mean complex Gaussian vector. The two-pairing formula is recovered only after the additional condition $\Pi = 0$ has been established.

The augmented covariance

$$\Gamma = \begin{pmatrix} C & \Pi \\ \Pi^* & C^* \end{pmatrix}$$

must be positive semidefinite. Complete whitening can be implemented as a real-linear transformation on the support of Γ . Equivalently, one can retain the original response coordinates and subtract $W(C, \Pi)$. We use the latter representation because it keeps the physical tangent labels visible. Exact null directions are removed according to a rank tolerance fixed by the second-moment audit, never by minimizing the fourth-order residual.

D. Pair-split estimator and jackknife uncertainty

Let $r = 1, \dots, R$ label the declared statistical units. For a fixed-Hamiltonian tangent-panel calculation, each r is one complete panel treated as exchangeable conditional on the common base Hamiltonian, not one channel or one matrix entry. Write $T^{(r)}$ for its physical four-point trace and let $W^{(r,s)}$ denote the three-pairing Wick tensor whose two second-moment factors are taken from distinct units $r \neq s$. The pair-split estimator is

$$\hat{\mathcal{K}} = \frac{1}{R} \sum_{r=1}^R T^{(r)} - \frac{1}{R(R-1)} \sum_{r \neq s} W^{(r,s)}.$$

Using distinct units in the quadratic products removes the leading same-sample plug-in bias. The ordered-pair form above is equivalent to an unordered-pair average after symmetrizing $W^{(r,s)}$ and $W^{(s,r)}$.

For a registered scalar functional $\widehat{\theta} = \mathcal{L}(\widehat{\mathcal{K}})$, delete unit r , recompute the entire estimator as $\widehat{\theta}_{(-r)}$, and form the delete-one pseudovalues

$$p_r = R\widehat{\theta} - (R-1)\widehat{\theta}_{(-r)}, \quad \bar{p} = R^{-1} \sum_r p_r.$$

The reported jackknife standard error is

$$\text{SE}_J = \left[\frac{1}{R(R-1)} \sum_{r=1}^R (p_r - \bar{p})^2 \right]^{1/2}.$$

This uncertainty is conditional on exchangeability of the declared units. A panel jackknife at one Hamiltonian measures variation within that tangent construction; it cannot be relabeled as disorder or Hamiltonian uncertainty.

E. Exact finite-population centering and frozen inference split

The corrected complete-cumulant calculation uses 24 registered panels in each case. Centering is performed before the split by an exactly evaluated mean of the finite tangent population, not by a sample average over the same panels used for inference. For a Moore–Read system with N_ϕ guiding centers, the population consists of the N_ϕ^2 torus anchors. Linearity of the protected response gives

$$\overline{X}_a = X \left(N_\phi^{-2} \sum_{x,y} G_{x,y} \right),$$

and the uniform average of the traceless guiding-density generators vanishes. For the local lattice-supercharge protocol, we enumerate the ordered coordinate marginals generated by the registered randomized ordering and average their projector responses. This accounts for the nonzero population mean of that discrete tangent construction.

After subtracting $\overline{X}_a$, panels $0, \dots, 11$ determine the Hermitian channel Gram matrix. Its positive inverse square root is frozen and applied to the disjoint inference panels $12, \dots, 23$. The training Gram after whitening is required to agree with the 2×2 identity, and singular or ill-conditioned channel support fails closed. Interchanging the two halves defines a reverse split, but its result is descriptive: it is not pooled and does not carry a claim gate.

Let $\widehat{\theta}$ be the fixed scalar contraction of the centered complete cumulant and s_J its delete-one jackknife error from the twelve inference panels. We form $t = \widehat{\theta}/s_J$ and compare it with

a Student- t_{11} reference. Since five primary cases were registered, the one-sided family level 0.05 is implemented by the Bonferroni per-case level 0.01. A case passes only if its one-sided p value is below 0.01 and the corresponding one-sided lower bound is positive. This gate is conditional on exchangeability of complete panels at the fixed Hamiltonian and does not estimate Hamiltonian-to-Hamiltonian fluctuations.

F. Form-factor normalization and finite-rank references

For a Hermitian matrix with eigenvalues f_i , our raw and connected form factors are

$$K_{\text{raw}}(\tau) = D^{-1} \mathbb{E} \left| \sum_{i=1}^D e^{-i\tau f_i} \right|^2, \quad K_c(\tau) = K_{\text{raw}}(\tau) - D^{-1} \left| \mathbb{E} \sum_{i=1}^D e^{-i\tau f_i} \right|^2.$$

For exactly degenerate shifted energies, all $f_i = 0$, so the disconnected piece exhausts the raw form factor and $K_c = 0$. For curvature eigenvalues, the same subtraction removes the one-point density and retains the connected two-level correlation.

When the null is a finite complex-Jacobi point process on $[-1, 1]$, its continuous joint density has the form

$$p(\{f_i\}) \propto \prod_{i < j} (f_i - f_j)^2 \prod_i (1 - f_i)^\alpha (1 + f_i)^\beta.$$

The finite-rank kernel is constructed from orthonormal Jacobi polynomials, and the connected form factor is the Fourier transform of the corresponding connected two-point function. If the compression dimensions force exact boundary atoms, those deterministic eigenvalues are separated before the continuous kernel is compared. This reference tests a specified finite- D compression ensemble; agreement over a registered interval is not an asymptotic universality statement.

II. NUMERICAL METHODS, AUDITS, AND EXTENDED RESULTS

This section specifies how the protected spaces and tangent responses are constructed. Numerical values, including ranks, gaps, residuals, basis dimensions, and production status, enter through the generated v13 macros and tables. The procedures below define those quantities and the gates applied to them; they do not infer a missing gate from a visually narrow spectrum.

A. Common kernel and response audit

For a factorized parent, we assemble or apply the rectangular constraint map C and solve for its right kernel. For a cohomological parent, we construct the charge-resolved Hodge Laplacian and solve its zero eigenspace. For a stabilizer code, the protected rank follows from binary symplectic elimination rather than from allocating the exponentially large state space. In every dense or sparse eigensolve the accepted projector frame U_P must satisfy

$$\|U_P^\dagger U_P - I_D\| < \epsilon_{\text{orth}}, \quad \|HU_P - E_0 U_P\|_F < \epsilon_{\text{ker}}.$$

The observed rank must equal the independently calculated protected rank. We then locate the first eigenvalue outside the target fiber and require a positive gap. An internal bandwidth smaller than a numerical tolerance is reported as a residual check; it is not substituted for the algebraic kernel condition.

Response vectors are obtained by applying the complement inverse to $Q(\partial_a H)P$. Dense cases use the stored positive-energy eigenbasis. Factorized large cases solve shifted or projected linear systems without forming $H = C^\dagger C$ densely. Each response is checked for target leakage, linear-system residual, and stability under the registered solver shift. Tangent panels are normalized with their own two-point Gram matrix before a four-channel quantity is evaluated.

B. Kapit–Mueller and continuum V_0 parents

The Kapit–Mueller single-particle hopping is assembled on a square magnetic torus, including its magnetic-periodic winding phases. The lowest isolated band is diagonalized, and its orbital frame is used to map onsite boson pairs into the projected N -particle Fock basis. The columns of the resulting pair constraint factor encode the pair annihilators A_R of the V_0 parent. Kernel vectors are computed from the factor, then checked against the explicitly applied positive parent. Quasihole ranks are computed from cyclic $(1, 2)$ admissible configurations before the eigensolve and are compared with the observed nullity.

For the independent continuum check, N bosons occupy n_ϕ Landau-gauge orbitals on a square torus. Magnetic translates of the periodic Gaussian give a factorized V_0 pair map from the N -particle basis to the $(N - 2)$ -particle basis. This code path shares neither the Kapit–Mueller hopping builder nor its orbital projection. Agreement of the protected count

therefore checks the two-body clustering mechanism across two microscopic realizations, whereas disagreement of higher response statistics remains an allowed parent dependence.

Density tangents are diagonal one-body operators in real space. Bond and current tangents are respectively the Hermitian and anti-Hermitian quadratures of a hopping link. For the exactness-preserving constraint/generator transport, the one-body generator is lifted to the fixed- N Fock space and applied to both the constraint factor and the projector. We verify $S_a = 0$ along the path and compare the transported kernel with a direct kernel solve. The v6 protected-generator results belong to exact transport. Only this calculation supports finite-path and holonomy conclusions. For twist-bundle calculations, every mesh edge stores the overlap singular values needed for polar parallel transport; a mesh is rejected if the target rank or external gap changes at any twist.

The v2/v3 Laughlin fixed-parent local-potential panels are evaluated separately. Their structured or random local potentials are applied only at the accepted exact parent, and their response is computed from $Q(\partial_a H)P$ for the isolated rank- D cluster. These probes generally do not preserve exact degeneracy away from the base point and therefore do not establish a finite exact path. Their admissible conclusions are the base-point curvature spectrum, covariance, and four-channel statistics.

C. Moore–Read coherent-state constraint map

In the N -boson occupation basis of N_ϕ torus orbitals, each coherent annihilator $b_x = \sum_j \phi_{xj} b_j$ is cubed and divided by $\sqrt{3!}$. Stacking C_x over an $N_\phi \times N_\phi$ magnetic-translation orbit gives a sparse map from the N -particle Hilbert space to the $(N-3)$ -particle space. The parent is applied as two sparse multiplications, $v \mapsto C^\dagger(Cv)$. The expected rank is computed from cyclic $(2, 2)$ admissible configurations before the solver is run.

The opened cases use periodic boundary conditions and the fixed-four- quasihole relation $N_\phi = N + 2$. The stored kernel audit contains the basis dimension, constraint shape and nonzero count, observed and expected ranks, zero-band width, first positive eigenvalue, residual norm, and orthonormality error. The v9 Moore–Read guiding-center panels are exactness-preserving generator transports. For each one-body generator G_a , the differentiated constraint-intertwiner identity gives $(\partial_a H)P = -H\Gamma_N(G_a)P$ and $X_a = Q\Gamma_N(G_a)P$ in the main-text Kato convention. The historical v9 routine stores instead $\tilde{X}_a = -Q\Gamma_N(G_a)P =$

$-X_a$, because its solver variable is $R_Q(\partial_a H)P$. It checks the parent-action residual of $\tilde{X}_a$, complement leakage, and the condition `fiber_tangent_zero`, which is the numerical $S_a = 0$ test. The registered quadratic and fourth-order contractions are invariant under this common sign, but the distinction is retained for response-factor provenance. The v9 panels are therefore not generic splitting probes. Larger staged cases remain marked pending until the same gates are passed; resource estimates are not converted into scientific results.

D. Random cubic supercharge

For N complex fermion modes, a charge- r basis is stored as ordered bit strings of Hamming weight r . Fermionic parity signs are included when the cubic annihilation operator $Q(C) = \sum_{i < j < k} C_{ijk} \psi_i \psi_j \psi_k$ is assembled. The charge-restricted Hamiltonian is the corresponding block of $Q^\dagger Q + Q Q^\dagger$. A coupling realization is accepted only if nilpotency, Hermiticity, the expected cohomology rank, a positive complement, kernel residual, and orthogonality all pass independently.

Couplings are normalized complex three-forms. The registered sizes and central or adjacent charge sectors are stored with independent realization seeds. Sparse tangent panels vary a fixed subset of C_{ijk} ; isotropic panels vary all cubic couplings with the same norm. The response is evaluated both from the direct Hamiltonian derivative and from its exact and coexact Hodge branches. We require the two branches to be orthogonal and their sum to reproduce the direct response. The random-supercharge calculation has independent coupling realizations for its registered separable null tests, but the v13 evidence table leaves complete nonseparable covariance closure untested.

E. Lattice supercharge on cycle unions

For $G = \bigsqcup_{s=1}^m C_6$, an occupation bit string is retained only when no graph edge has both endpoints occupied. The charge- f basis thus enumerates the size- f faces of the independence complex. The sparse map $Q_f : \mathcal{H}_f \rightarrow \mathcal{H}_{f+1}$ includes the Jordan–Wigner parity associated with the canonical vertex order. Direct multiplication verifies $Q_{f+1} Q_f = 0$. In sector $f = 2m$ we diagonalize

$$H_f = Q_{f-1} Q_{f-1}^\dagger + Q_f^\dagger Q_f.$$

The first 2^m eigenvectors must exhaust the numerical kernel, and the next eigenvalue must be positive.

Complex site couplings are normalized separately on each six-site ring. Candidate local or isotropic tangent vectors are projected orthogonally to every component-wise rescaling direction, then orthonormalized. This removes the trivial deformation that merely rescales one Hodge block. Derivatives of both terms in H_f give exact and coexact response branches. We compare their sum with the direct resolvent derivative and check mutual orthogonality, target leakage, and nonzero branch weights. The statistical unit in the complete-covariance calculation is one full fixed-base tangent panel. It is not an independent graph or Hamiltonian realization. For the corrected v13 calculation, the mean of the local-panel population is obtained by exact enumeration of the ordered coordinate-channel marginals. The first twelve panels then determine only the channel whitener, while the remaining twelve panels enter the primary complete-cumulant estimator.

F. X-cube binary-symplectic construction

Edges of the periodic cubic lattice are indexed by position and one of three directions. A Pauli operator is represented by a binary row $(\mathbf{x}|\mathbf{z})$. The symplectic inner product verifies that every cube- X row commutes with every vertex-plane- Z row. Gaussian elimination over $\mathbb{F}_2$ gives the stabilizer rank, logical-qubit count, and hence the protected rank without constructing a 2^{3L^3} -dimensional vector.

For positive coefficient reweighting, the stabilizer table and common $+1$ eigenspace are identical at every parameter value, so $\partial_a P = 0$ follows algebraically. For the moving-projector control, two local Pauli generators are selected whose product differs from a stabilizer only by phase. Their support, anticommutation, detectability, exact isospectrality, and gap preservation are checked in the binary representation. A small dense Pauli realization independently checks the sign of the curvature. Production sizes use the algebraic proof and never allocate the ground-space matrix.

G. Checkerboard low-energy cluster

The checkerboard control is built from a two-sublattice short-range hopping matrix on a torus. Complex nearest-neighbor and real longer-range hoppings produce an isolated lower Chern band. At every boundary twist, the lowest single-particle band frame is recomputed and the onsite boson contact interaction is projected into its complete fixed- N occupation basis. This is the same projected-contact convention used for the growing-rank Kapit–Mueller comparison, but with an independently constructed hopping matrix. Boundary twists are inserted as Peierls phases on seam-crossing bonds before the projection. The tracked multiplet is selected by dimension and continuity over the twist mesh. Acceptance requires its internal width to remain smaller than its external separation, but the width is not required to vanish.

Bond-magnitude, bond-phase, and zero-mean onsite disorder define the deformation. Neighboring subspaces are compared by singular values and polar parallel transport before their curvature increments are formed. Because the cluster is only quasi-degenerate, its energy statistics remain meaningful and are reported separately from the exactly degenerate parents. No checkerboard result is used to establish a common-kernel, cohomology, or stabilizer claim.
